Hospital Operational & Revenue Analysis

Identifying Workload Imbalance, Utilization Gaps & Financial Risks

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TOOLS: POWER BI | POWER QUERY | DAX

Problem Statement

Nowadays hospitals manage multiple departments, doctors, appointments, and financial operations simultaneously.
However, hospitals often struggle to:

- Monitor revenue and profit performance clearly
- Balance workload between senior, mid-level, and junior doctors
- Track department capacity and actual utilization
- Control increasing no-show rates
- Reduce patient waiting time and improve satisfaction

Due to these mismanagement operational inefficiencies may arise, leading to reduced performance, patient satisfaction and long-term sustainability risks.

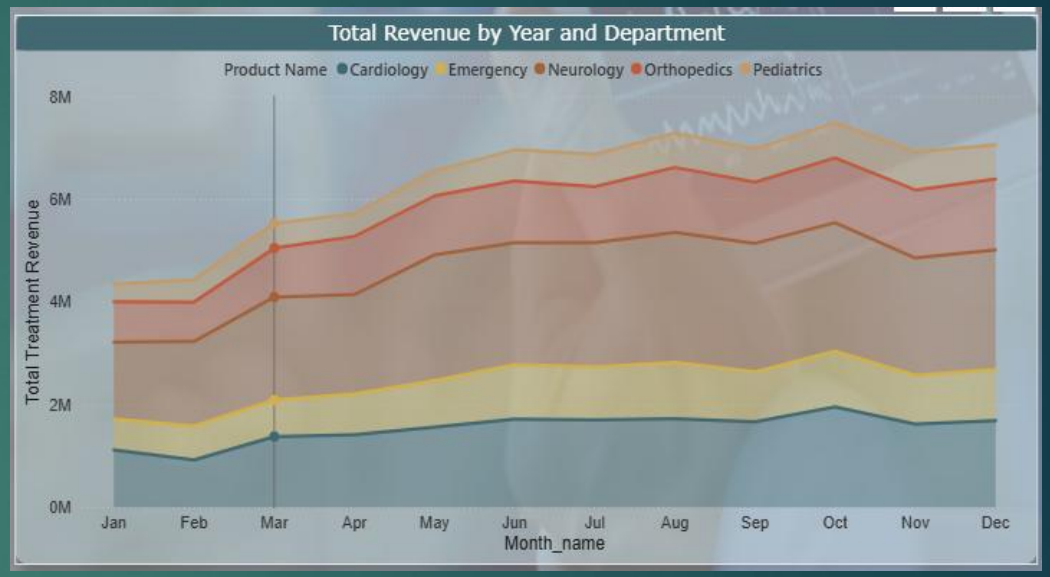
This project aims to analyze hospital operational and financial data and build an interactive Power BI dashboard that gives clear visibility into revenue, utilization, doctor performance, no-show trends, and waiting time — enabling better and data-driven decision-making.

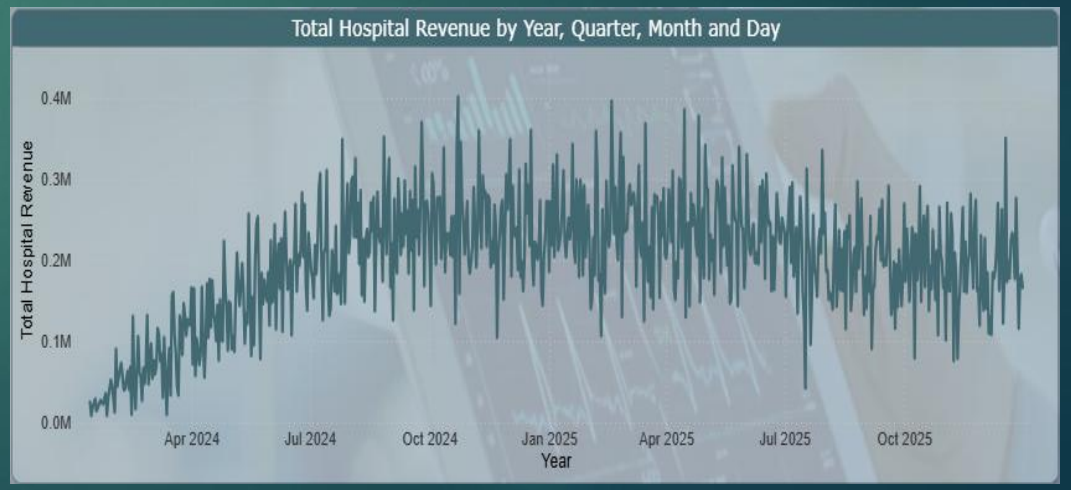
Business Objectives

- Monitor hospital revenue and profit performance
- Analyze doctor workload across seniority levels
- Evaluate department capacity and actual utilization
- Track no-show rates and patient waiting time trends
- Identify operational gaps and improvement opportunities

Key Insights – Revenue Performance

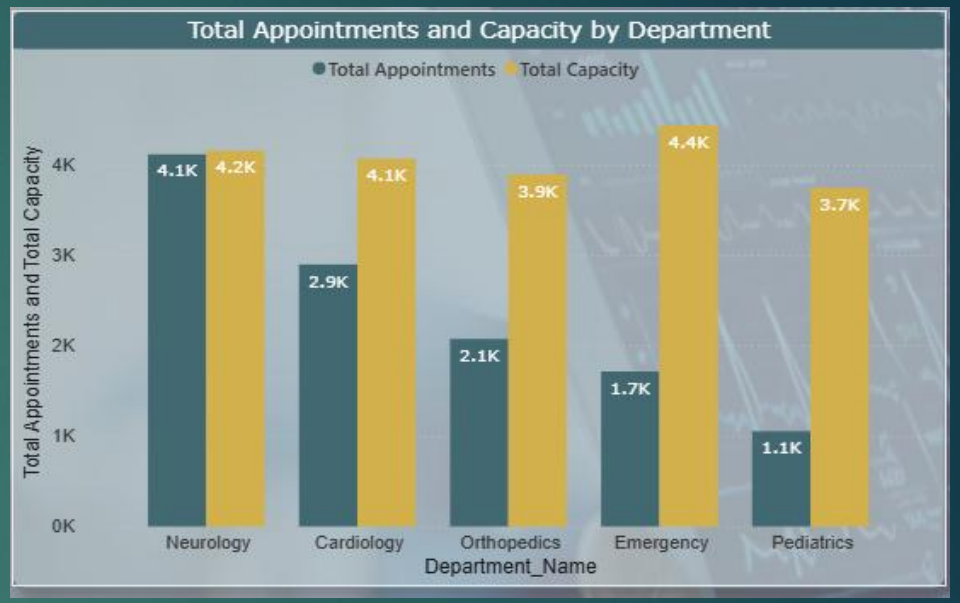
- Revenue shows slight decline in recent months not a high difference but needs improvement
- Cardiology and Neurology are the main department which drives revenue, shows that revenue is not evenly distributed across departments
- Treatment revenue slightly exceeds Appointment revenue, indicating dependence on treatments

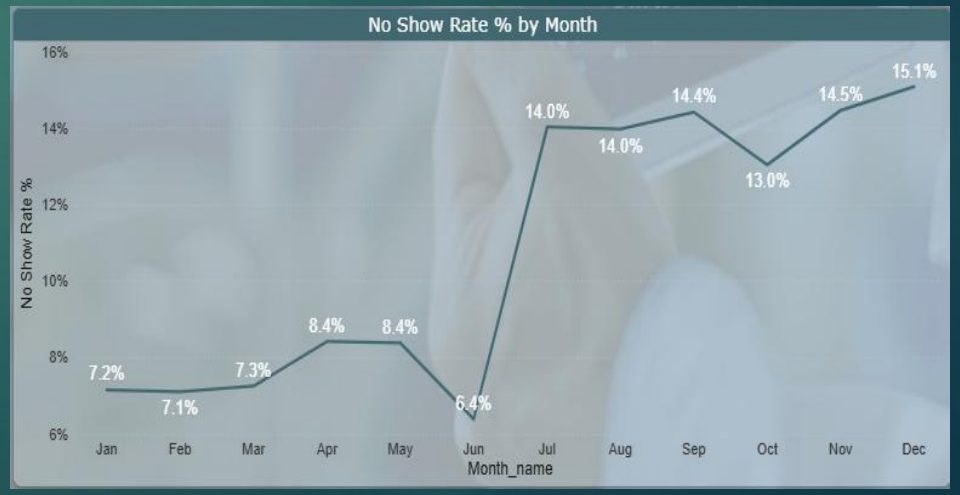




Key Insights – Operational Performance

- Department utilization levels for some departments are below optimal capacity but neurology is slightly over used.
- A small group of doctors (e.g., Navya, Jhon, Vikram) are overutilized, while several others remain underutilized, indicating clear workload imbalance across the hospital
- No-show rates have increased in recent months, showing patient dissatisfaction.
- Rising patient waiting time may negatively impact satisfaction levels
- Senior doctors generate the majority of hospital revenue and profit increasing dependency risk





Business Impact

- Improve workload distribution between senior and junior doctors by pairing senior doctors with junior doctors for mentoring, skill transfer and increasing trust.
- Improve department capacity and improve under utilized departments by identifying tasks in the overloaded department that can be done by the underutilised department.
- Improve patient experience by reducing waiting time effectively
- Improve No show patients by managing wait time and an option to reschedule
- Research shows that patient no show ranging from 6% to 23%, cost the healthcare industry over $150 billion annually. we can reduce this by implementing multiple automated reminder and easy rescheduling options

Future Scope

- Integrate real-time hospital data to monitor performance continuously
- Develop predictive models to forecast patient demand and appointment trends
- Build a no-show prediction system and also improve scheduling options
- Implement workload forecasting to better balance doctor allocation
- Expand the dashboard to support long-term strategic planning such as hiring more doctors or expanding departments based on performance trends.

Conclusion

This project shows how Power BI can transform raw hospital operational data into meaningful and actionable insights.

The Hospital Operations Dashboard provides a complete view of revenue performance, doctor workload, department utilization, no-show trends, and patient waiting time.

By using this dashboard, healthcare management teams can:

- Make faster and data-driven decisions
- Reduce operational inefficiencies
- Balance doctor workload effectively
- Improve patient experience
- Strengthen long-term operational sustainability

Thank You